# Module–4

1. Let S be the set defined by $S = \{z \in C : |z| = 1\}$, where C is the set of all complex numbers. Show that S forms a commutative group under usual multiplication of complex numbers.

$$S = \{z \in C : |z| = 1\}$$

① Closure

let $z_1, z_2 \in S$

Then $|z_1| = 1$, $|z_2| = 1$

$|z_1 z_2| = |z_1||z_2| = 1$

$\therefore z_1 z_2 \in S$

therefore z is closed under multiplication.

② Associativity

Complex multiplication is associative.

$z_1(z_2 z_3) = (z_1 z_2) z_3$

$\forall\ z_1, z_2, z_3 \in S$

③ Identity $a \cdot e = a$

Now, $z \cdot e = z$

$\therefore e = 1$.

$\therefore$ identity element exists and that is $e = 1$

④ Inverse since $z = 1$

$z\bar{z} = |z|^2 = 1$

$\therefore z^{-1} = \bar{z}$

Also, $|\bar{z}| = |z| = 1$

$\therefore \bar{z} \in S$

⑤ Commutative.

$z_1 z_2 = z_2 z_1$

$\therefore (S, \cdot)$ is an Abelian gr. Proved

2. Show that all roots of the equation $x^4 = 1$ form an Abelian group under multiplication.

$$G = \{x \in \mathbb{C} : x^4 = 1\}$$

$\therefore G = \{1, -1, i, -i\}$

| X | 1 | -1 | i | -i |
|---|---|---|---|---|
| 1 | 1 | -1 | i | -1 |
| -1 | -1 | 1 | -i | i |
| i | i | -i | -1 | i |
| -i | -i | i | i | -1 |

① Closure

every product $\in G$

hence closure holds.

② Associativity

Multiplication of complex numbers is associative.

$(ab)c = a(bc)\ \ \forall\ a, b, c \in G$

③ Identity: $a * e = e * a = a$

here $z \cdot e = z$

$1 \cdot e = 1 \rightarrow e = 1$

& $1 \in G$

④ Inverse

$a * a^{-1} = e$

$= 1$

$1 \cdot 1^{-1} = 1$

$-1 \cdot (-1)^{-1} = 1$

$i \cdot i^{-1} = 1$

$(-i) \cdot (-i)^{-1} = 1$

⑤ Commutative

$a*b = b*a$

Now, for $a = i,\ b = 1$ : $i.1 = 1.i$

$a = -1,\ b = i$ : $-1.i = i.-1$

$a = i,\ b = -i$ : $i.-i = -i.i$

$\therefore$ $(G, \cdot)$ is an Abelian group.

3. Show that the set of rational numbers other than one, $Q' = Q - \{1\}$ forms a group under the binary operation * defined by $a * b = a + b - ab, \forall\, a, b \in Q'$.

$a*b = a+b-ab$

① Closure

$a, b \in Q'$

$a+b-ab \in Q'$

② Associativity

$(a*b)*c$

$= (a+b-ab)*c$

$= a+b-ab+c-c(a+b-ab)$

$= a+b+c-ab-bc-ac+abc$

$a*(b*c)$

$= a*(b+c-bc)$

$= a+b+c-bc-a(b+c-bc)$

$= a+b+c-ab-bc-ca+abc$

$\therefore (a*b)*c = a*(b*c)$

③ Identity

$a*e = a$

$\therefore a+e-ae = a$

$\therefore e-ae = 0$

$e = \dfrac{0}{1-a} = 0 \in Q'$

④ Inverse

$a*a^{-1} = e$

$\therefore a*a^{-1} = 0$

or $a+a^{-1}-a.a^{-1} = 0$

or, $a^{-1}-a.a^{-1} = -a$

or, $a^{-1} = -\dfrac{a}{1-a} \in Q'$

⑤ Commutative

$a*b = a+b-ab$

$b*a = b+a-ab$

so,

$a*b = b*a$

$\therefore (Q', *)$ is an Abelian group.

4. Show that $G = \{a + b\sqrt{2}, \forall\, a, b\ \in Q\}$ form a group with respect to multiplication.

$0 = 0+0\sqrt{2}$ and $0$ has no multiplicative inverse.

$\therefore$ the set should be $G = \{a+b\sqrt{2}, \forall\, a,b \in Q,\ a+b\sqrt{2} \neq 0\}$

① Closure

let $\left.\begin{array}{l} x = a+b\sqrt{2} \\ y = c+d\sqrt{2} \end{array}\right\} \in G$

$\therefore xy = (a+b\sqrt{2})(c+d\sqrt{2})$

$= ac+2bd+(ad+bc)\sqrt{2}$

here $a, c, b, d \in Q \wedge (ad+bc) \in Q$

$\therefore xy \in G$

② Associativity

Multiplication of real numbers are associative.

$(ab)c = a(bc)$

$\forall\, a, b, c \in G$

③ Identity

$a \cdot e = a$

$\therefore e = 1$

& $1 \in G$

because $1 = 1 + 0\sqrt{2} \in G$

④ Inverse

$a \cdot a^{-1} = e$

$a \cdot a^{-1} = 1$

$\therefore a^{-1} = \frac{1}{a}$

Suppose

$a = x + y\sqrt{2} \neq 0$

$\therefore a^{-1} = \frac{1}{x + y\sqrt{2}}$

$= \frac{x - y\sqrt{2}}{x^2 - 2y^2}$

Since, $x, y \in Q$, denominator $x^2 - 2y^2 \neq 0$

unless $(x = y = 0)$

[already excluded]

Therefore $x^{-1} = \frac{x}{x^2 - 2y^2} - \frac{y}{x^2 - 2y^2}\sqrt{2}$

which is again of the form $P + Q\sqrt{2}$

with $P, Q \in Q$

hence $x^{-1} \in G$

⑤ Commutativity

Multiplication of real Numbers are commutative

$ab = ba, \ \forall\ a, b \in G$

$\therefore (G, \cdot)$ is an Abelian group. (Proved)

5. Prove that the set M of all 2 × 2 non-singular real matrices form a group under matrix multiplication.

$M = \{A : A$ is a $2 \times 2$ non singular real matrix$\}$

① Closure

let, $A, B \in M$

$|A| \neq 0$, $|B| \neq 0$ [$\because$ non-singular]

Now,

$|AB| = |A||B| \neq 0$

$\therefore AB \in M$

$\therefore$ M is closed under Matrix Multiplication.

② Associativity

$(AB)C = A(BC) \ \forall\ A, B, C \in M$

Matrix Multiplication is Associative.

③ Identity

$A \cdot e = A$

$e = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\quad |e| \neq 0$

& $A \cdot e = e \cdot A = A, \ \forall\ A \in M$

$\therefore e$ is an identity element.

④ Inverse

Let, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M$

$$A^{-1} = \frac{Adj(A)}{|A|} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$(ad-bc \neq 0)$

Since, $|A^{-1}| = \frac{1}{|A|} \neq 0$

$\therefore \quad A^{-1} \in M$.

$\therefore$ All the 4 properties satisfied -

- Closure
- Associativity
- Identity
- Inverse

$\therefore$ $(M, \cdot)$ is a group under Matrix Multiplication.

(Proved)

6. Let a be an element of a group $(G, \circ)$ of order n. Then $a^m = e$, identity element if and only if n is a divisor of m.

Let, $O(a) = n$ (order of the element $a$ is $n$) then

$$a^m = e \iff n \mid m$$

Proof

We know that, $O(a) = n$ means: $n = \min\{k > 0 : a^k = e\}$

7. $O(y) = 20$, $O(y^8) = ?$

$n = 20$, $k = 8$

$\therefore O(y^8) = \frac{20}{\gcd(20,8)} = \frac{20}{4} = 5$ Ans

Theorem,

if $O(a) = n$,

$O(a^k) = \frac{n}{\gcd(n,k)}$

8. Theorem

A non empty subset $H$ of group $(G, \circ)$ is a subgroup of $G$ if and only if

$a, b \in H \Rightarrow a \circ b^{-1} \in H$

We have to prove that $H$ is a subgroup.

1. Show the identity $e \in H$

Since $H$ is non empty, take any $a \in H$.

Put $b = a$

$\therefore aa^{-1} \in H$

$\therefore e \in H$

2. Show that inverse are in $H$

We ~~know~~ $e \in H$

take any $a \in H$

using the condn with $a = e$ and $b = a$, $ea^{-1} \in H$

$a^{-1} \in H$

3. Show closure

take $a, b \in H$

we know $b^{-1} \in H$ (from S-2)

Now with the condn $a$ & $b^{-1}$: $aa^{-1} \in H$

$a(b^{-1})^{-1} \in H$

$ab \in H$

$\therefore H$ is closed under the op.

$\therefore H$ is subgroup of $G \rightarrow \boxed{H \leq G}$ Proved

9. Intersectn of 2 subgrps is a subgr. $\rightarrow$ Proof

Let, $H_1$ and $H_2$ be subgroups of G.

We have to show $H_1 \cap H_2$ is a subgr.

Take any $a, b \in (H_1 \cap H_2)$ then $\left.\begin{matrix} a, b \in H_1 \\ a, b \in H_2 \end{matrix}\right\}$

Since $H_1$ & $H_2$ are subgroups.

$ab^{-1} \in H_1 \wedge ab^{-1} \in H_2$

$\therefore ab^{-1} \in (H_1 \cap H_2)$ $\therefore$ Proved $(H_1 \cap H_2)$ is a subgroup (Proved)

Union of 2 subgr.s may not be a subgr.

Consider group $(\mathbb{Z}, +)$

Let, $H_1 = \{\ldots, -4, -2, 0, 2, 4, \ldots\} \Rightarrow$ even int

$H_2 = \{\ldots 6, 3, 0, 3, 6, \ldots\} \Rightarrow$ Multiples of 3.

both are subgroups of $\mathbb{Z}$

Now, $2 \in H_1$, $3 \in H_2$

$\therefore 2, 3 \in (H_1 \cup H_2)$

but $2 + 3 = 5 \notin (H_1 \cup H_2)$

So the union is not closed.

$\therefore (H_1 \cup H_2)$ is not a subgroup. (Proved)

10.

**Definition of Cyclic Group**

A group $G$ is called a **cyclic group** if there exists an element $a \in G$ such that every element of $G$ can be written as a power of $a$.

$$G = \langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$$

The element $a$ is called a **generator** of the group.

Every cyclic gr. is abelian.

Let $G = \langle a \rangle$ be a cyclic gr.

Take any 2 elements $x, y \in G$

Since $G$ is cyclic there exist integers $m, n$ such that

$$x = a^m, \; y = a^n$$

Now, $xy = a^m a^n = a^{m+n}$

Also, $yx = a^n a^m = a^{n+m}$

Since int addition is commutative $m+n = n+m$

$\therefore \; xy = a^{m+n} = a^{n+m} = yx$

Hence, $xy = yx \; \forall x, y \in G$

Thus $G$ is abelian $\therefore$ Every cyclic gr. is abelian.

11. $G = \{1, 2, 3, 4, 5, 6\}$

Check powers of 3 mod 7

$3^1 \equiv 3 \pmod 7$
$3^2 \equiv 2 \pmod 7$
$3^3 \equiv 6 \pmod 7$
$3^4 \equiv 3 \cdot 6 \equiv 18 \equiv 4 \pmod 7$
$3^5 \equiv 2 \cdot 6 \equiv 12 \equiv 5 \pmod 7$
$3^6 \equiv 6 \cdot 6 \equiv 36 \equiv 1 \pmod 7$

Thus,
$\langle 3 \rangle = \{1, 2, 3, 4, 5, 6\} = G$

$\therefore G$ is a cyclic gr.

Find all Generators.

A generator must have order 6.

numbers before reaching to 1

check $\boxed{2}$ :
$2^1 = 2 \equiv 2 \pmod 7$
$2^2 = 4 \equiv 4 \pmod 7$
$2^3 = 8 \equiv 1 \pmod 7$

$\therefore o(2) = 3$
$\therefore$ Not a generator.

$\boxed{3}$ : $3^1 = 3 \equiv 3 \pmod 7$
$3^2 = 9 \equiv 2 \pmod 7$
$\vdots$
$3^6 = 36 \equiv 1 \pmod 7$

$\therefore o(3) = 6$
$\therefore$ it is a generator.

$\boxed{4}$ : $4^1 = 4 \equiv 4 \pmod 7$
$4^2 = 16 \equiv 2 \pmod 7$
$4^3 = 4 \times 2 = 8 \equiv 1 \pmod 7$

$\therefore O(4) = 3$
$\therefore$ not a generator.

$\boxed{5}$ : $5^1 = 5 \equiv 5 \pmod 7$
$5^2 = 25 \equiv 4 \pmod 7$
$5^3 = 20 \equiv 6 \pmod 7$
$5^4 = 5^2 \cdot 5^2 = 25 \cdot 25 = 5 \times 5 \equiv 25 \equiv 4 \pmod 7$
$5^5 = 4 \cdot 5 = 20 \equiv 6 \pmod 7$
$5^6 = 6 \times 6 = 36 \equiv 1 \pmod 7$

$\therefore O(5) = 6$
$\therefore$ generator.

$\boxed{6}$ : $6^1 = 6 \equiv 6 \pmod 7$
$6^2 = 36 \equiv 1 \pmod 7$

$\therefore O(6) = 2$
$\therefore$ not a generator.

$\therefore$ generators are $\rightarrow$ $\boxed{3, 5}$ Ans

12. $(Z_6, +) \rightarrow$ modulo

$Z_6 = \{0, 1, 2, 3, 4, 5\}$

Show $Z_6$ cyclic.

$1 = 1 \equiv 1 \pmod 6$
$1+1 = 2 \equiv 2 \pmod 6$
$1+1+1 = 3 \equiv 3 \pmod 6$
$1+1+1+1 = 4 \equiv 4 \pmod 6$
$1+1+1+1+1 = 5 \equiv 5 \pmod 6$
$1+1+1+1+1+1 = 6 \equiv 0 \pmod 6$

$\rightarrow \therefore \langle 1 \rangle = \{0, 1, 2, 3, 4, 5\} = Z_6$

$\therefore Z_6$ is cyclic.
(Proved)

Find all generators — (For addition $n.a = 0$).

$\boxed{1} \Rightarrow$ $1 = 1 \pmod 6$
$1+1 = 2 \pmod 6$
$\vdots$
$1+1+1+1+1+1 = 6 \equiv 0 \pmod 6$

$\rightarrow \therefore O(1) = 6$
$\therefore$ generator.

$\boxed{2} \Rightarrow$

$1.2 = 2 \equiv 2 \pmod 6$
$2.2 = 4 \equiv 4 \pmod 6$
$3.2 = 6 \equiv 0 \pmod 6$

$\therefore O(2) = 3$
$\therefore$ not generator.

$\boxed{3} \Rightarrow$

$1.3 = 3 \equiv 3 \pmod 6$
$2.3 = 6 \equiv 0 \pmod 6$

$\therefore O(3) = 2$
$\therefore$ not generator

$\boxed{4} \Rightarrow$

$1.4 = 4 \equiv 4 \pmod 6$
$2.4 = 8 \equiv 2 \pmod 6$
$3.4 = 12 \equiv 0 \pmod 6$

$\therefore O(4) = 3$
$\therefore$ not generator.

$\boxed{5} \Rightarrow$

$1.5 = 5 \equiv 5 \pmod 6$
$2.5 = 10 \equiv 4 \pmod 6$
$3.5 = 15 \equiv 3 \pmod 6$
$4.5 = 20 \equiv 2 \pmod 6$
$5.5 = 25 \equiv 1 \pmod 6$
$6.5 = 30 \equiv 0 \pmod 6$

$\therefore O(5) = 6$
$\therefore$ generator.

$\therefore$ generators $\rightarrow \boxed{5, 1}$ Ans

13. $Z_9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

# Show that $Z_9$ is cyclic.

$1 \times 1 = 1 \equiv 1 \pmod 9$
$2 \times 1 = 2 \equiv 2 \pmod 9$
$3 \times 1 = 3 \equiv 3 \pmod 9$
$4 \times 1 = 4 \equiv 4 \pmod 9$
$5 \times 1 = 5 \equiv 5 \pmod 9$
$6 \times 1 = 6 \equiv 6 \pmod 9$
$7 \times 1 = 7 \equiv 7 \pmod 9$
$8 \times 1 = 8 \equiv 8 \pmod 9$
$9 \times 1 = 9 \equiv 0 \pmod 9$

$\therefore \langle a \rangle = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$
$= Z_9$

$\therefore Z_9$ is cyclic.

# Find generators

$\boxed{1} \Rightarrow$

$1 \times 1 = 1 \equiv 1 \pmod 9$
$2 \times 1 = 2 \equiv 2 \pmod 9$
$\vdots$
$9 \times 1 = 9 \equiv 0 \pmod 9$

$\rightarrow \therefore O(1) = 9$ ✓
$\therefore$ generator.

$\boxed{2} \Rightarrow$

$1 \times 2 = 2 \equiv 2 \pmod 9$
$2 \times 2 = 4 \equiv 4 \pmod 9$
$\vdots$
$5 \times 2 = 10 \equiv 1 \pmod 9$
$6 \times 2 = 12 \equiv 3 \pmod 9$
$\vdots$
$9 \times 2 = 18 \equiv 0 \pmod 9$

$\rightarrow \therefore O(2) = 9$ ✓
$\therefore$ generator.

$\boxed{3} \Rightarrow$

$1 \times 3 = 3 \equiv 3 \pmod 9$
$\vdots$
$3 \times 3 = 9 \equiv 0 \pmod 9$

$\rightarrow \therefore O(3) = 3$
$\therefore$ not a generator. ✗

$\boxed{4} \Rightarrow$

$1 \times 4 = 4 \equiv 4 \pmod 9$
$\vdots$
$9 \times 4 = 36 \equiv 0 \pmod 9$

$\rightarrow \therefore O(4) = 9$ ✓
$\therefore$ generator.

$\boxed{5} \Rightarrow$

$1 \times 5 = 5 \equiv 5 \pmod 9$
$\vdots$
$9 \times 5 = 45 \equiv 0 \pmod 9$

$\rightarrow \therefore O(5) = 9$ ✓
$\therefore$ generator.

$\boxed{6} \Rightarrow$

$1 \times 6 = 6 \equiv 6 \pmod 9$
$\vdots$
$3 \times 6 = 18 \equiv 0 \pmod 9$

$\rightarrow \therefore O(6) = 3$
$\therefore$ Not generator. ✗

$\boxed{7} \Rightarrow$ $1 \times 7 = 7 \equiv 7 \pmod 9$
$\vdots$
$9 \times 7 = 63 \equiv 0 \pmod 9$

$\rightarrow \therefore O(7) = 9$
$\therefore$ generator ✓

$\boxed{8} \Rightarrow$ $1 \times 8 = 8 \equiv 8 \pmod 9$
$9 \times 8 = 72 \equiv 0 \pmod 9$

$\rightarrow \therefore O(8) = 9$
$\therefore$ generator ✓

$\therefore$ generators $= \{1, 2, 4, 5, 7, 8\}$ Ans

14. $G = \{e, a, a^2, a^3, a^4, a^5, a^6, a^7\}$ , $a^8 = e$

This is a cyclic gr. generated by a and has order $= 8$.

$O(a) = 8$

$\therefore O(a^k) = \dfrac{8}{\gcd(8,k)}$

$O(e) = O(a^8) = \dfrac{8}{\gcd(8,8)} = 1$

$O(a) = \dfrac{8}{\gcd(8,1)} = 8$

$O(a^2) = \dfrac{8}{\gcd(8,2)} = \dfrac{8}{2} = 4$

$O(a^3) = \dfrac{8}{\gcd(8,3)} = \dfrac{8}{1} = 8$

$O(a^4) = \dfrac{8}{\gcd(8,4)} = \dfrac{8}{4} = 2$

$O(a^5) = \dfrac{8}{\gcd(8,5)} = 8$

$O(a^6) = \dfrac{8}{\gcd(8,6)} = \dfrac{8}{2} = 4$

$O(a^7) = \dfrac{8}{\gcd(8,7)} = 8$

15. Let H be a proper subgr. of a gr. G of order 6.

we have $|G| = 6$.

By Joseph Lewis Lagrange Theorem $\rightarrow |H| \mid |G|$.

$|H|$ divides $|G|$.

Since, the divisors of 6 are — 1, 2, 3, 6.

and H is a proper subgr. ($H \neq G$)

we must have $|H| = 1, 2, 3$.

Case 1 $|H| = 1$ :

H contains only 1 element.

then $|H| = \{e\}$

this is the trivial cyclic gr.

case 2 $|H| = 2$

Let $H = \{e, a\}$ $\because a \neq e$ $\therefore O(a) = 2$

$\therefore H = \langle a \rangle$

So, $H$ is cyclic

case 3 $|H| = 3$.

Let, $a \in H$, $a \neq e$.

Since 3 is prime

$O(a) = 3$

Thus

$H = \{e, a, a^2\} = \langle a \rangle$

$\therefore H$ is cyclic.

$\therefore$ Every proper subgr. of a gr. of order 6 is cyclic.

(Proved)

17. (a) $f = (1,2,3)(1,2)$

3 cycle len → $3-1 = 2$ tran. → even

2 len → $2-1 = 1$ → odd

$2 + 1 = \boxed{3}$

Total = odd.

$\therefore f$ is an odd permutation.

Determine whether a permutation is even or odd

- $n$ cycle length → $(n-1)$ Transpositions
  - odd → odd permutation
  - even → even permutation

(b) $g = (12345)(123)(45)$

$4 + 2 + 1 = 7$ odd

18. inverse of permutations.

(a) $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$

$\sigma^{-1} = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$

(top row sorting) Ans

(b) $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$

$\sigma^{-1} = \begin{pmatrix} 2 & 3 & 1 & 5 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}$

Ans

19. $f : (\mathbb{Z}, +) \to (\mathbb{R}^+, \cdot)$, $f(x) = 4^x$

Homomorphism

for any $m, n \in \mathbb{Z}$

$f(m, n) = 4^{m+n} = 4^m \cdot 4^n = f(m)\, f(n)$

$\therefore$ $f$ is a homomorphism.

One to One

suppose $f(m) = f(n)$

$\therefore 4^m = 4^n$

$\therefore m = n$

$\therefore$ $f$ is injective (one to one)

$\therefore$ $f$ is monomorphism.

Check onto

The codomain is $\mathbb{R}^+$, but numbers like 2, 3, 5 are not of the form $4^x$ for any integer $x$.

$\therefore$ $f$ is not surjective. $\therefore$ $f$ is not an isomorphism.